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Measures for assessing the data freshness in Open Data portals

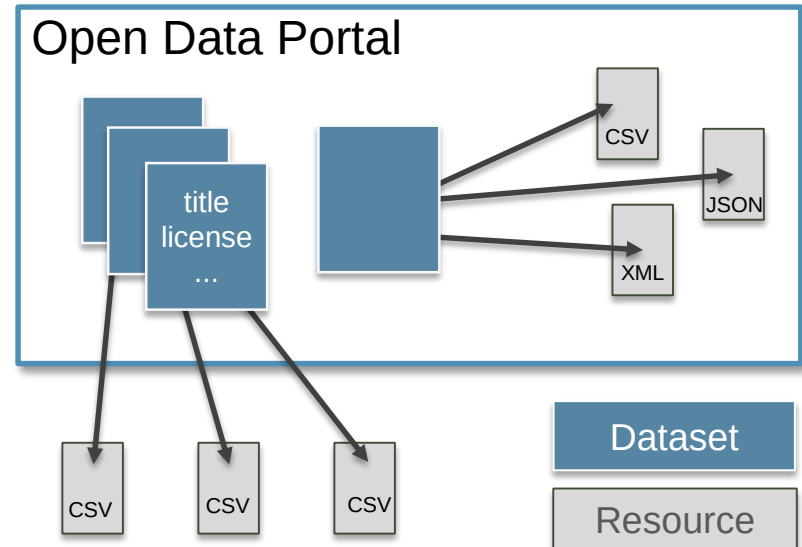
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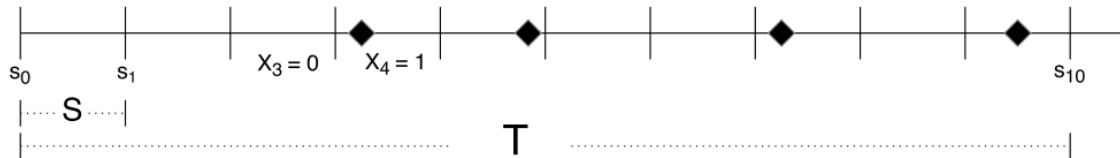
- How up-to-date are resources in Open Data portals?
- Required information for such a metric:
 - **Change history** of documents in portals
- Challenge:
 1. Collect available change history
 2. Estimated next change time to assess up-to-dateness
- Two scenarios:
 - *Portal provider*: wants to add freshness measure to metadata
 - *Data consumer*: updating of application, DB, etc..

Open Data Portals

- Single point of access
- Local and external resources
- Meta data
 - Title
 - **Modification date**
 - ...
- Typical software:



Sources of change information in OD portals



■ **Push-based** history:

- Data provider push change information to portal
 - If *local*, by uploading new version
 - If *external*, by updating a specific metadata field

```
last_modified: "2013-09-25T00:00:00"
```

■ **Pull-based** history:

■ **Age sampling:**

- Access to latest change time of a resource
(i.e., last-modified timestamp in *HTTP Header*)

```
Last-Modified: Mon, 04 Nov 2013 13:00:08 GMT  
ETag: "21096456bff7d72268dc99b3bf082565"
```

■ **Comparison sampling:**

- Detect changes by monitoring and comparing the resources

Open Data Portal Watch

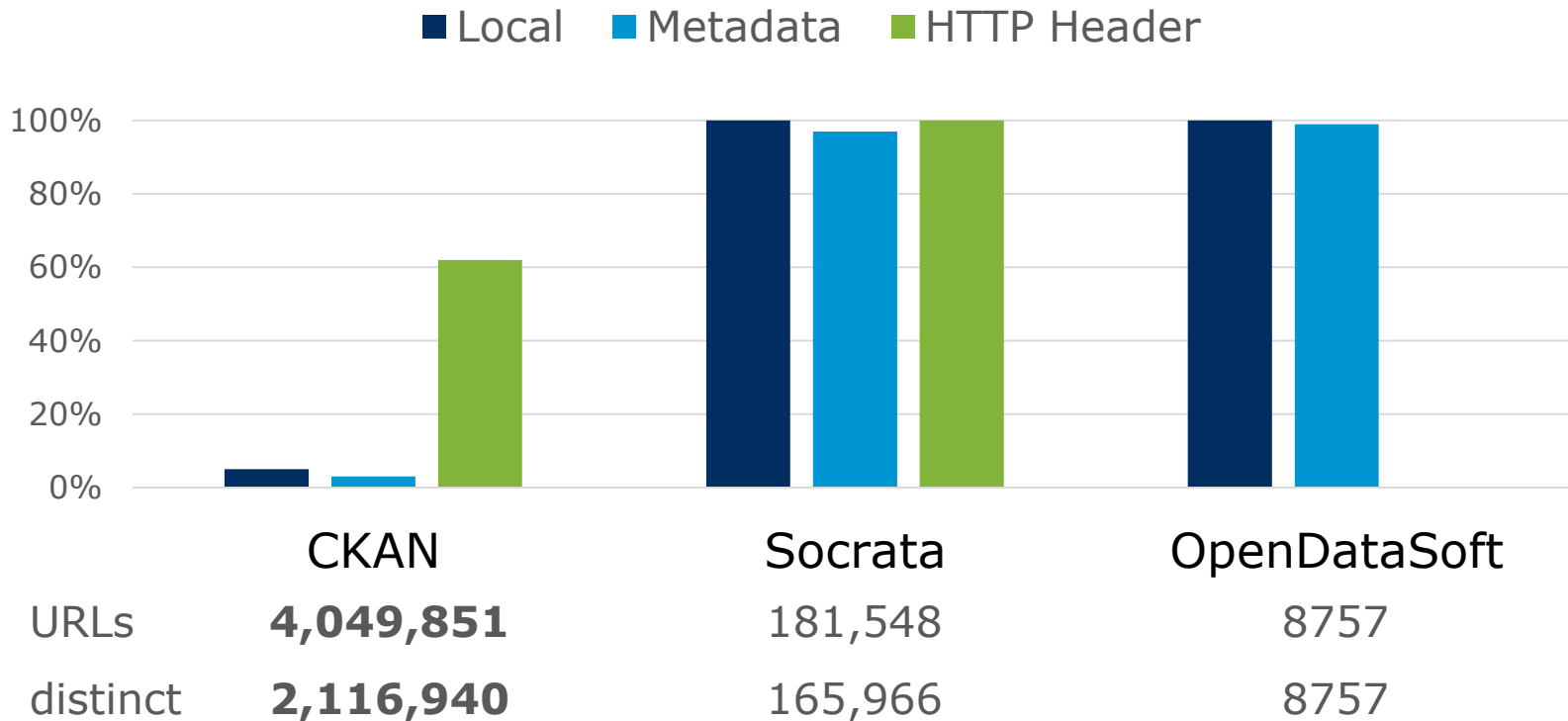
<http://data.wu.ac.at/portalwatch/>

- Periodically monitoring over 260 Open Data portals
- Metadata quality assessment
 - Uniform handling of metadata (using DCAT mapping)
- Evolution tracking & archiving
 - Meta data
 - Data

The screenshot shows the 'Open Data Portal Watch' website. The header is blue with the site name and navigation links: Info, Portals, API, Quality Measures, and About. Below the header, a blue banner states 'Portal List containing 261 portals'. A search bar with the placeholder 'Filter' is present. Below the search bar, there are sorting options: 'SORT BY URL', 'SORT BY ISO3', 'SORT BY SOFTWARE', 'SORT BY #DATASETS', and 'SORT BY #RESOURCES'. Two portal entries are visible:

- http://opendata.paris.fr.opendatasoft.com**
FRA OpenDataSoft 804 resources 199 datasets
[SEE PORTAL DETAILS](#)
- http://datos.argentina.gob.ar/**
ARG CKAN 11 resources 8 datasets
[SEE PORTAL DETAILS](#)

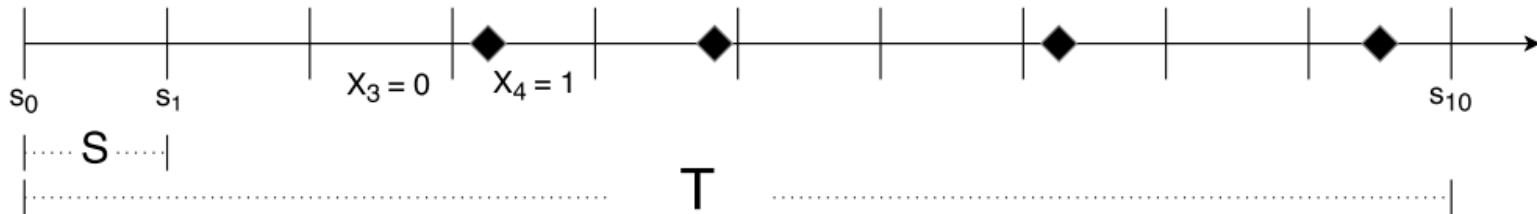
Available change information



- *CKAN*: age- and comparison-sampling required
- *Socrata & OpenDataSoft*: push-based possible

- Evaluating three change estimation heuristics:
 - *Poisson process*
 - Cho and Garcia-Molina (2003) propose Poisson process model to estimate updates in the context of Web sites
 - *Markov chain approach*
 - Umbrich et al. (2015) use Markov chains to schedule next crawl times for URLs based on previous observed changes
 - *Empirical distribution*
 - Build empirical distribution of changes based on intervals

Estimation of next updates (cont'd)



■ Age sampling

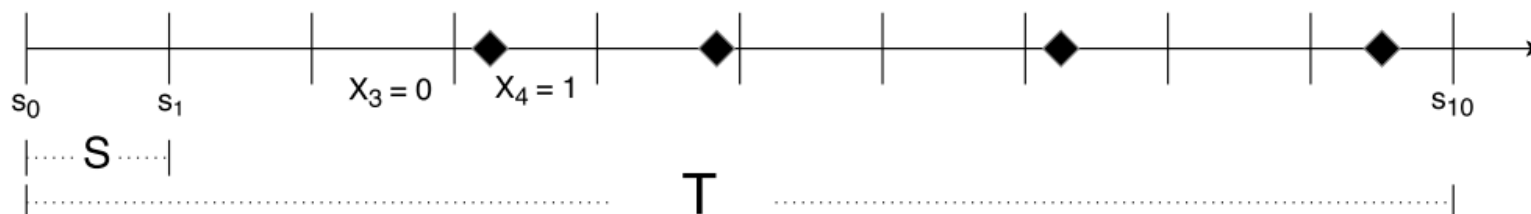
■ *Poisson distribution*

- X/T ($= \frac{\text{number of changes}}{\text{monitoring period}}$) as estimator for Poisson parameter
- Compute next change time by considering p -quantiles

■ *Empirical distribution*

- Use intervals between the observed last-modified times
- p -quantiles of empirical distribution

Estimation of next updates (cont'd)



■ Comparison sampling

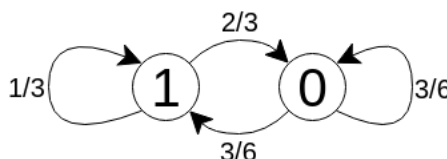
- Only binary information/states available:

| 0 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 0 | 1 |

■ Markov chain approach

- Probability of next change based on previous state, e.g.:

$i \setminus i+1$	1	0	TOTAL
0	3	3	6
1	1	2	3



$$P(1|0) = 3/6$$

Evaluation Summary

- Controlled environment:
 - Evaluation using revision histories of Wikipedia articles
 - 1562 randomly Wiki articles with >3 years history and >30 revisions
 - Wiki change history does not follow Poisson distribution

- Different confidence values:

- *For fixed p , we report the ratio of successfully predicted updates*

- Conclusion:

- *Markov chain approach* best for comparison-based sampling
 - *Empirical distribution* best for push-based and age-based sampling

COMPARISON SAMPLING RESULTS.						
Estimator	<i>All</i>		<i>Regular</i>		<i>Irregular</i>	
p = 0.7 S = 10d						
C _{EmpDist}	0.59	40d	0.66	40d	0.60	90d
C _{ChoNaive}	0.67	36d	0.67	35d	0.63	83d
C _{ChoImpr}	0.66	35d	0.62	34d	0.61	82d
C _{UmbMarkov}	0.51	42d	0.76	41d	0.73	96d
p = 0.7 S = 50d						
C _{EmpDist}	0.54	40d	0.57	40d	0.57	84d
C _{ChoNaive}	0.65	37d	0.36	40d	0.63	78d
C _{ChoImpr}	0.27	43d	0.31	36d	0.47	76d
C _{UmbMarkov}	0.58	39d	0.59	40d	0.68	82d
p = 0.9 S = 10d						
C _{EmpDist}	0.81	66d	0.87	70d	0.80	145d
C _{ChoNaive}	0.71	38d	0.70	37d	0.67	85d
C _{ChoImpr}	0.57	36d	0.66	35d	0.60	83d
C _{UmbMarkov}	0.88	84d	0.94	85d	0.90	184d

Thank you for your attention

- *Goal*
 - Data Freshness estimation in Open Data
- *Challenge*
 - Collecting change history (push vs pull)
- *Approach*
 - Estimators for different scenarios
 - Empirical evaluation



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